An Improved Online Testing Technique For Reversible Circuits

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***Abstract*—The emerging technology of reversible circuits offers a potential solution to the synthesis of ultra low-power quantum computing systems. A reversible circuit can be envisaged as a cascade of reversible gates only, such as Toffoli gate, which has two components: k control bits and a target bit (k-CNOT), $k \geq 1$. While analyzing testability issues in a reversible circuit, the missing-gate fault model is often used for modeling physical defects in k-CNOT gates. In this paper, we propose online design-for-testability (DFT) technique. The proposed method is an improved version of an earlier work by Kole et. al. Our method yields less overhead in terms of quantum cost as compared to the original approach. The method is advantageous for circuits where consecutive gates occur frequently with the same set of controls.**

***Index Terms*—Missing-gate fault, quantum computing, reversible logic, test generation**

I. Introduction

Rising concerns for energy loss have become a fundamental problem in modern nano-scale CMOS logic circuits. According to Landauer's principle [1], a non-reversible circuit causes irreversibility of information processing with which, energy loss is associated. Reversible logic, on the other hand, provides a mechanism for designing information lossless circuits, and may provide a solution to energy reduction for futuristic technology [1]–[3]. It also provides a foundation for designing quantum computers [4], [5].

Classical logic gates such as AND, OR, NAND, NOR, or XOR used in digital design are not reversible, and hence, they cannot be used to design reversible circuits. In order to design reversible circuits, only reversible gates can be used such as controlled-not (CNOT) gate proposed by Feynman [6], Toffoli [7], or Fredkin [8] gates, among others.

Several researchers [9]–[12] have studied the problem of fault modelling and testing of reversible logic circuits. While the stuck-at fault model is widely used for conventional logic circuits, newer models such as single missing-gate fault (SMGF), repeated-gate fault (RGF), partial missing-gate fault (PMGF), or multiple missing gate fault (MMGF), have been found to be more suitable for representing physical failures or defects in reversible circuits implemented with quantum technologies [9], [11], [12]. An online testing technique was discussed in [13].

In this paper, we propose a method to reduce the number of additional gates required in the design [13]. We have compared the proposed technique with the original technique and have shown that they outperform the existing techniques with respect to quantum cost overhead for a number of benchmark reversible circuits.

The paper is organized as follows. Section II presents the basic definitions associated with reversible logic. Section III presents the original technique [13] proposed by Kole et. al. Section IV describes our proposed modified designs. Section V depicts the experimental results and comparison, and section VI contains the conclusion.

II. Preliminaries

A. Reversible Logic

A Boolean function is reversible if there exists a bijective mapping between inputs and outputs. In a reversible function, the input can be produced back from the output. In others words, the output pattern has to be a permutation of the input pattern. To satisfy all these characteristics, the number of output lines should be equal to the number of input lines.

B. Reversible gates

The basic CNOT-type reversible gates used for synthesis are the following:
(i) $(1 \times 1) NOT(x_1 \rightarrow x_1')$
(ii) (2×2) controlled NOT (CNOT):$(x_1, x_2) \rightarrow (x_1, x_1 \oplus x_2)$
(iii) (3×3) Toffoli gate $(x_1, x_2, x_3) \rightarrow (x_1, x_2, x_1 x_2 \oplus x_3)$.
The standard graphic symbol for the above mentioned three gates are shown in Figure 1. A generalized Toffoli gate, also known as Multiple Control Toffoli (MCT) gate [8], has a set of control inputs *C*, a target input set *T*, and has the form *TOF(C;T)*, where $C = (x_{i1}, x_{i2}, ..., x_{ik})$, $T = \{x_j\}$ and $C \cap T = \phi$. It maps an input vector $(x_1^0, x_2^0, ..., x_n^0)$ to $(x_1^0, x_2^0, ..., x_{j-1}^0, x_j^0 \oplus (x_{i1}^0 . x_{i2}^0 ..... x_{ik}^0), x_{j+1}^0, ..., x_n^0)$. Thus, a NOT gate is $TOF(x_j)$, i.e., a generalized Toffoli gate which has no controls as shown in Figure 1(a). The CNOT gate is $TOF(x_i, x_j)$, i.e., a generalized Toffoli gate with one control bit [6] as shown in Figure 1(b); this is also known as Feynman gate. The simple (3×3) Toffoli gate is a generalized Toffoli gate with two controls [6] as shown in Figure 1(c). A *k*-CNOT gate has *k* control inputs $x_1, x_2, ..., x_k$ and one target input t.

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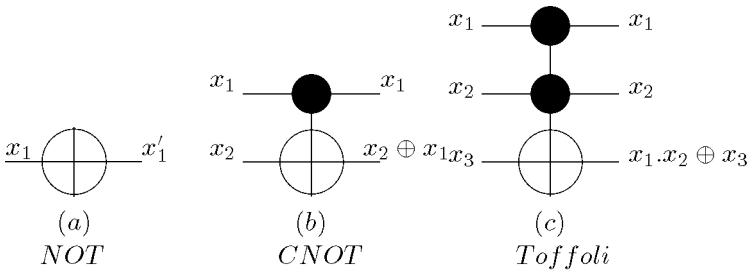

Fig. 1. Three basic reversible gates.

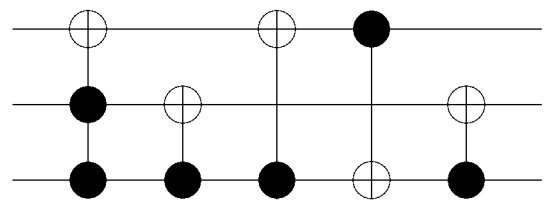

Fig. 2. A Reversible Circuit

TABLE I
QUANTUM COST OF TOFFOLI GATE WITH n CONTROL LINES

# controls	0	1	2	3	4	5	6	7	8
Quantum Cost ($Q_c(k)$)	1	1	5	14	20	32	44	64	76

It maps the input vector $(x_1, x_2, ..., x_k, t)$ to the output vector $(x_1, x_2, ..., x_k, t \oplus x_1.x_2...x_k)$.

C. Quantum Cost Analysis

The quantum cost of a gate G is the number of elementary quantum operations required to realize the function given by G. Table I shows the required quantum cost $Q_c(k)$ in the realization of Multiple Control Toffoli (MCT) gate with k controls [14]. An MCT gate with $(k+1)$ lines has k control lines.

D. Reversible Logic Circuit

A reversible logic circuit RC of size n with depth d is a cascade of d reversible gates over the set of input lines $X = \{x_1, x_2, \cdots, x_n\}$, i.e. $RC = g_0 \cdot g_1 \cdot g_2 \cdots g_{d-1}$ where each g_i represent i^{th} reversible gate of the circuit RC.
A logic circuit is said to be reversible if it is fan-out free, acyclic, and consists of only reversible gates, which themselves implement reversible functions; such gates need to be specially designed, e.g., Toffoli gates. All the conventional logic gates except NOT, are irreversible. An example of a reversible logic circuit is shown in Figure 2. The quantum cost of a reversible logic circuit is the summation of the quantum cost of all the reversible gates in the circuit.

E. Garbage

A garbage output of a reversible circuit is an output whose functionality is not a part of design specification. The garbage outputs are the extra outputs those are required to realize the given function and to ensure reversibility.

F. Modeling of faults and their detection

Several fault models for k-CNOT based reversible circuits were introduced earlier. In this paper we follow the model proposed in [9] as described below.

A *single missing gate fault* (SMGF) is the removal of one k-CNOT gate from the circuit. Consider a circuit shown in Figure 3(a). Figure 3(b) shows a faulty circuit with a SMGF fault where one gate has disappeared from the area marked by the dotted box. If we apply (1,1,1,1) at the input of the fault-free circuit in Figure 3(a), the normal output would be (1,1,1,0), but in the presence of SMGF fault, the output for the faulty circuit in Figure 3(b) will be (1,1,0,1). A SMGF fault is detected by a test which produces 1 at all controls of the missing gate.

The *repeated-gate fault* (RGF) model occurs when a gate is replaced by several instances of the same gate in the circuit. When a gate is replaced by odd (even) number of of times, the fault is undetectable (detectable). All detectable RGF faults are equivalent and is detectable by a test which produces 1 at all controls of the repeated gate. Figure 3(c) shows a circuit with a RGF fault. This fault is detected by vector (1,1,1,1).

The *multiple missing-gate fault* (MMGF) model is the disappearance of a set of consecutive gates from the circuit. All MMGFs are not detectable. Figure 3(d) shows an MMGF in which the position of the consecutive missing gates are shown by a box. This fault is detected by the vector (0,1,1,1).

A partial missing-gate fault (PMGF) is the disappearance of m of k controls in a k-CNOT gate, thus transforming it into a $(k-m)$-CNOT gate. The number m is called the order of the PMGF. A PMGF fault is detected by a test which produces 0 at the missing control and 1 at all other control. Figure 3(e) shows an PMGF with a box enclosing the missing control. This fault is detected by vector (0,1,1,1). In a k-CNOT gate, there are k first order PMGF. To test these k first order PMGF, k distinct tests are required. These k test can detect all higher order PMGF. Thus to detect all PMGF in a k-CNOT gate, we require k tests.

III. ONLINE DESIGN PROPOSED BY DIPAK KOLE ET. AL. [13]

Kole et. al. proposed an online testing technique in [13] for detecting SMGF and PMGF faults in reversible circuits. In this technique, a additional line C_1 is introduced in the circuit. For every gate $TOF(C, T)$ where C represents the set of controls and T is the target line , some additional gates are appended. The additional gates comprises of a set of three gates TOF(T, C_1), TOF(T, C_1), and TOF(C,C_1) as shown in Figure 4. These

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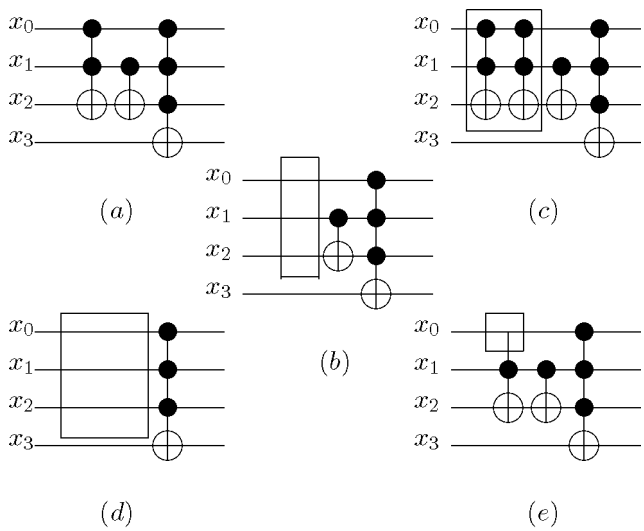

Fig. 3. Missing gate fault: (a) Original circuit (b) Single missing gate fault (SMGF) (c) Repeated gate fault (RGF) (d) Multiple missing gate fault (MMGF) (e) Partial missing gate fault (PMGF).

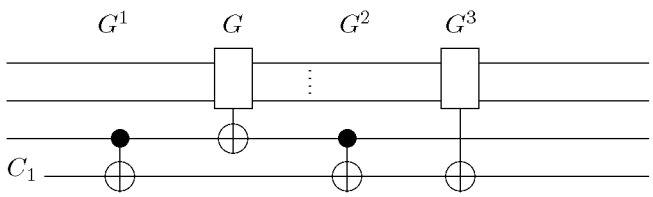

Fig. 4. Additional circuitry required for online testing for a gate.

additional gates are named as G^1, G^2, and G^3 as shown in the figure 4.

C_1 is maintained at value 0. G^1 and G^2 are 1-CNOT gates which produces the binary sum of input and output of target in G on C_1. Whenever 1 is input to all controls of G, the target flips the input to produce the output. G^1 and G^2 produces 1 in C_1. Aditional gate G^3 which has same set of controls as G, becomes active and restores the value of C_1 back to 0.

The additional gates functions to maintain value 0 at C_1. Therefore in a fault free circuit, 0 appears at C_1 for all inputs. Value 1 in C_1 indicates the presence of a fault in the circuit either in the original gates or in the additional gates. Thus by observing the value of the line C_1, it can be identified whether the circuit is faulty or fault-free.

Example III.1. *: Consider the reversible circuit as shown in Figure 2. The set of 3 additional gates are inserted for each gate following the method shown in Figure 4. The complete circuit after such insertion is shown in Figure 5.*

IV. PROPOSED DESIGN

In the paper, we have improved an earlier online testing technique given in [13]. Consider the circuit in Figure 2. Here the second and third gate has same set of controls, which indicates that at a particular instant of time, either both will become active or remain inactive.

When both gates are active simultaneously, the summation of binary sum produced by additional gates G^1, G^2 of both gates becomes 0. Hence additional gate G^3 can be omitted for both the gates.

Lemma 1. *Additional gates G_1 and G_2 for gate G_i and G_{i+1} maintains value 0 at C_1.*

Proof. Let input to target of $G_i = m$.
G_i^1 produces at $C_1 : m \oplus C_1 = m \oplus 0 = m$.
G_i^2 produces at C_1 when G is not activated (activated) : $m \oplus m(m') = 0(1)$.

Let input to target of $G_{i+1} = n$.
G_{i+1}^1 produces at $C_1 : n \oplus C_1 = n \oplus 0(1) = n(n')$.
G_{i+1}^2 produces at $C_1 : n(n') \oplus n(n') = 0$.

Thus G_1 and G_2 for gate G_i and G_{i+1} together maintains value 0 at C_1. □

Therefore, whenever there exist consequetive gates with set of control lines, the addition gates for each gate comprises of two gates G^1 and G^2. This procedure is well described in the algorithm DFT-1.

Algorithm 1 DFT-1

Input : A reversible circuit with size n and depth N
Output : Augmented circuit.
BEGIN
1. Introduce 1 Line C_1.
2. $i = 1$.
3. **while**($i \neq$N)
4. **if** (G_i, G_{i+1} has same set of controls)
5. Insert G_i^1 = TOF(Target(G_i), C_1)
6. Insert G_i
7. Insert G_i^2 = TOF(Target(G_i), C_1)
8. Insert G_{i+1}^1 = TOF(Target(G_{i+1}), C_1)
9. Insert G_{i+1}
10. Insert G_{i+1}^2 = TOF(Target(G_{i+1}), C_1)
11. $i = i + 2$
12. **else**
13. Insert G_i^1 = TOF(Target(G_i), C_1)
14. Insert G_i
15. Insert G_i^2 = TOF(Target(G_i), C_1)
16. Insert G_i^3 = TOF($Control(G_i)$, C_1)
17. $i = i + 1$
18. **endif**
19. **endwhile**
END

Target(G) and Control(G) are two functions used in the algorithms, which computes the target line and set of control line for gate G. The modified design for the circuit in Figure 2 is shown in Figure 6. In Figure 6, boxes are used to show the position of gates that has been removed.

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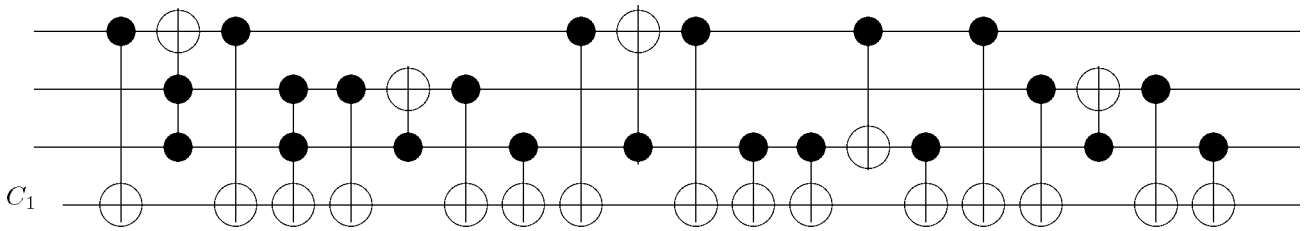

Fig. 5. Online testable design for the circuit in Fig. 2

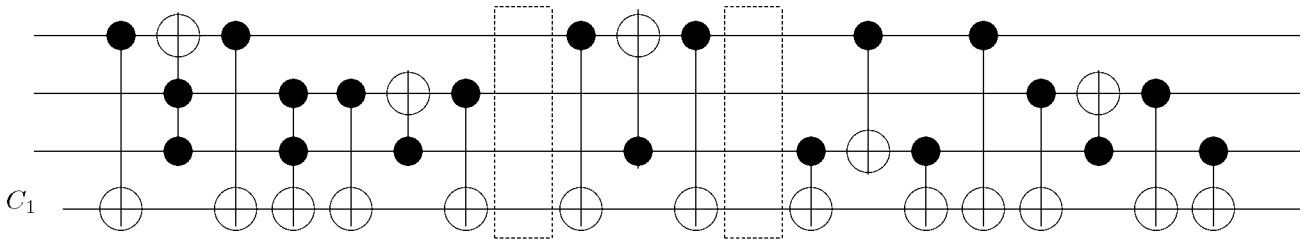

Fig. 6. Reduction of gates in online testable design for the circuit in Fig. 2

V. Experimental results

We have tested several reversible benchmark circuits, the results of which are shown in the Table. II. The table compares the proposed design with the work in [13] with respect to quantum cost overhead. The first 3 column of the table contains the circuit name, size and the depth of the corresponding circuits. The next two columns, which are divided into 2 subcolumn for comparing the proposed design and the DFT technique in [13] with respect to the quantum cost overhead. The last column contains the percentage decrement in quantum cost in our design with respect to [13] for the benchmark circuits.

From the table it can be seen that the proposed design incurs less quantum cost as compared to the approach in [13].

VI. Conclusion

We have proposed an improved version of the technique as proposed in [13]. Our design is found to be best suited for circuits in which pair of consecutive gates with same control set occurs frequently. For a number of benchmark examples, we have observed a significant reduction in quantum cost as compared to the original technique.

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TABLE II
COMPARISON WITH [13]

Circuit	size	depth	Quantum Cost	% increment in Q_c		% decrement in Q_c with respect to [13]
				[13]	Our Technique	
alu-v2_31	5	13	65	140 %	96.92%	30.77 %
C17_204	7	9	47	138.30%	104.26%	24.62 %
cmb_214	20	18	114	131.58%	87.72%	33.33 %
f2_232	8	19	107	135.51%	71.96%	46.9 %
rd53_251	8	27	123	143.90%	100%	30.5 %
5xp1_194	17	85	481	135.34%	86.28%	36.25 %
adr4_197	13	55	315	134.92%	105.71%	21.65 %
alpa_203	22	80	885	118.08%	41.02%	65.26 %
cm151a_211	28	33	275	124 %	102.911%	17.01 %
cm163a_213	29	39	235	133.192 %	88.94 %	33.23 %
cm42a_207	14	35	165	142.42%	56.36%	60.43 %
cu_219	25	40	308	125.97%	87.66%	30.41 %
dc1	11	39	209	137.32 %	56.94 %	58.54 %
dk17_224	21	49	419	123.39 %	76.13 %	38.30 %
dk27_225	18	24	98	148.98 %	104.08 %	30.14 %

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